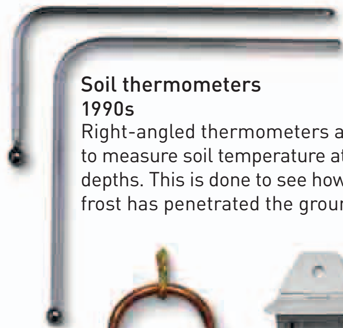
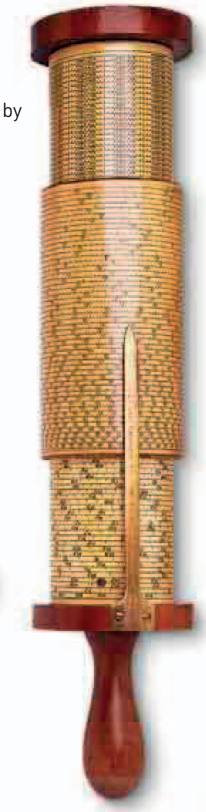
Glass thermometer
1700s
This beautiful thermometer was made by Italian glass blowers. It is filled with alcohol that expands and climbs the spiral when the temperature rises.



Soil thermometers
1990s
Right-angled thermometers are used to measure soil temperature at varying depths. This is done to see how deeply frost has penetrated the ground.



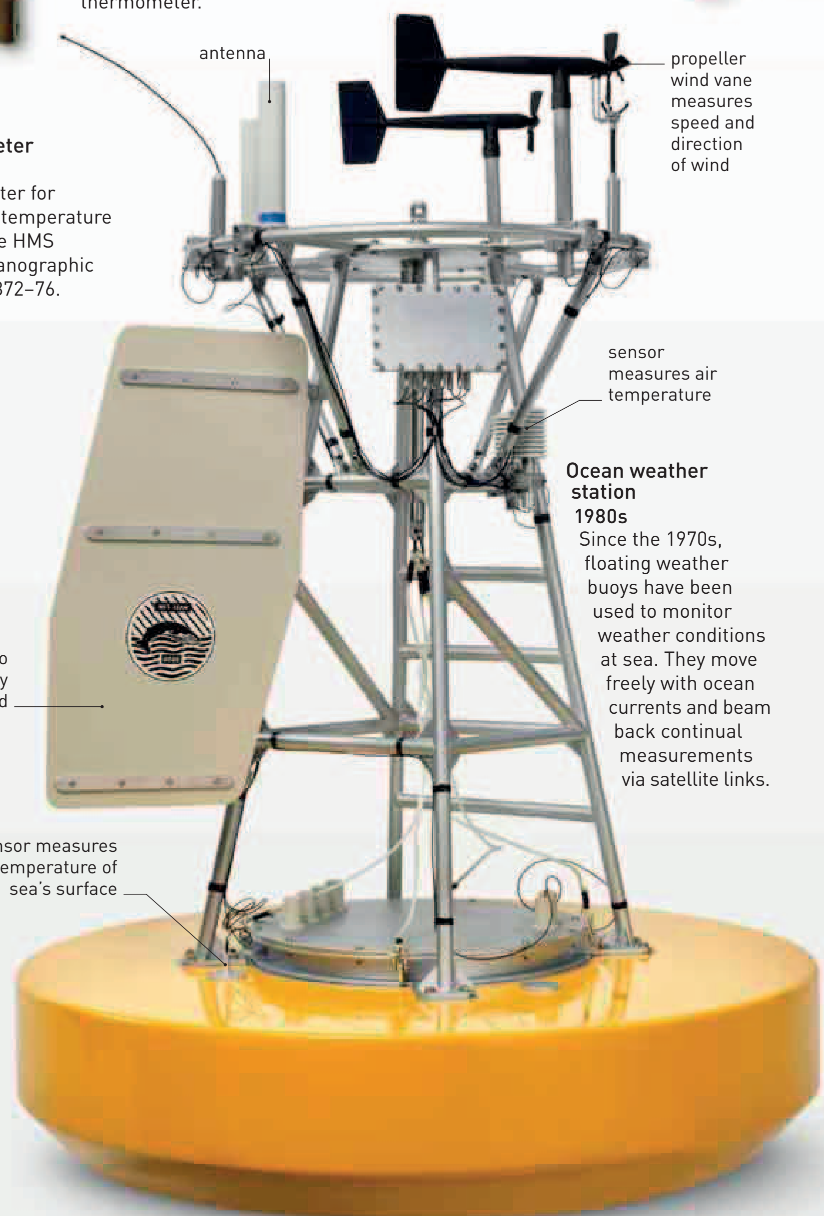
Weather calculator
1920s
British meteorologist Lewis Richardson helped develop numeric weather prediction by creating special calculators.

Thermometer
1990s
Maximum and minimum temperatures reached each day are recorded on either arm of a double-ended thermometer.

Cotton-reel thermometer
***c.*1855–77**
This desk-top combination instrument features a mercury thermometer and a compass.

Rain gauge
1980s
Rainfall can be recorded simply by the depth of water funnelled down inside a rain gauge, typically mounted 8 in (20 cm) above the ground to avoid splashes.

Sea thermometer
***c.*1870**
This thermometer for measuring sea temperature was used on the HMS *Challenger* oceanographic expedition of 1872–76.



Ocean weather station
1980s
Since the 1970s, floating weather buoys have been used to monitor weather conditions at sea. They move freely with ocean currents and beam back continual measurements via satellite links.

Hair hygrometer
***c.*1830**
This simple way of measuring humidity depended on the ability of a human hair to stretch in moist air and shrink in dry air in a regular and predictable way.

Hygrometer
1836
Evaporation causes cooling, so the temperature difference between two bulbs of a thermometer, one kept wet, one dry, can be compared to calculate humidity.